

Solution 1. Below is the integer program requested.

$$\begin{aligned}
& \text{Minimize} && c^\top x \\
& \text{subject to} && x \in \{0, 1\}^E \\
& && x(\delta(v)) \geq 1, \quad \text{for each } v \in V, \\
& && x(\delta(S)) \geq 1, \quad \text{for each } S \subseteq V \text{ such that } \emptyset \neq S \neq V
\end{aligned}$$

The above formulation uses the variable x , where each entry is associated with an edge $e \in E$. The entry of x correspondent to e is supposed to be equal to 1 if e is on a spanning tree, otherwise it will be 0.

Initially it is important to mention that a spanning tree it is a tree (acyclic) which the set of vertices of this tree is equal to V .

The first set of constraints $x(\delta(v)) \geq 1$ enforces that each vertex $v \in V$ is an end of at least one of the edges of the spanning tree. The second set of constraints $x(\delta(S)) \geq 1$ guarantee that the graph generated by set of edges that are correct answers for the problem is connected (technique used in class).

Given that, the set of edges that will have correspondent x entry equal to 1 is associated with a connected graph that has set of vertices equal to V . By associated graph, I mean the graph that has set of edges equal to set of edges that will have correspondent x entry equal to 1 and set of vertices equal to the set of all vertices that are an end of an edge that is an element of the set of edges. Given that, the only constrain that is not addressed yet is the fact that the associated graph need to be acyclic because it is a spanning tree. This is not a problem because, as we are minimizing $c^\top x$, a cycle will never be in our answer. This is true and trivial because, if there is a cycle on the associated graph, you can remove an edge that is on this cycle and the associated graph will continue to be connected, otherwise this edge would not be in a cycle by definition. Also, as every vertex in a cycle is an end of two distinct edges, the set of edges remains equal to V when you remove one edge from the cycle. Thus, $c^\top x$ gives the desired answer.

Solution 2. Below is the integer program requested.

$$\begin{aligned}
& \text{Maximize} && \mathbb{1}^\top e \\
& \text{subject to} && v \in \{0, 1\}^V, e \in \{0, 1\}^E \\
& && v_u + v_w + e_{uw} \leq 2, && \text{for each } \{u, w\} \in E, \\
& && v_u + v_w - e_{uw} \geq 0, && \text{for each } \{u, w\} \in E, \\
& && v_u - v_w - e_{uw} \leq 0, && \text{for each } \{u, w\} \in E, \\
& && v_w - v_u - e_{uw} \leq 0, && \text{for each } \{u, w\} \in E,
\end{aligned}$$

The above formulation uses the variable v , where v_u is supposed to be 1 if the vertex u is in a shore of the cut and 0 otherwise. It also uses the variable e , where the entry e_{uw} is supposed to be 1 if the edge $\{u, w\}$ is in the cut.

For this problem I wrote the constraints based on the valid scenarios. An edge $\{u, w\} \in E$, which has ends u and w , is in a cut if and only if exactly one of those vertices are in a shore of the cut. To clarify, my formulation accepts values of all possible shores and the cuts associated with each shore.

Now I will explain better each constraint. The top constraint $v_u + v_w + e_{uw} \leq 2$ invalidates the scenario where the 3 variables are 1. The second constraint $v_u + v_w - e_{uw} \geq 0$ does not allow v_u and v_w to be 0 and $e_{u,w}$ to be 1. The third and fourth constraints invalidate the scenario where only 1 vertex of the edge is in a shore but the edge is not in the cut. It is important to mention that the third and fourth constraints are symmetric and I just add both because edges are an unordered pair. More important than that, all valid scenarios satisfy all the constraints.

The sum of all entries of e that are equal to 1 is the number of edges in a cut. As we are maximizing this sum, we will have the maximum size of a cut of G .

Solution 3. I will prove that $\mathcal{H}^{\min}$ is a clutter and $B^{\min}$ is its blocker. From exercise 11 of assignment 0, we have that $\mathcal{H}^{\min} = (\mathcal{H}^{\min})^{\min}$, then $\mathcal{H}^{\min}$ is a clutter. For the other part, I need to prove that $b(\mathcal{H}^{\min}) = \mathcal{B}^{\min}$. Let $B(\mathcal{H}^{\min}) := (V, \{K \subseteq V : K \text{ is a vertex cover of } \mathcal{H}^{\min}\})$, which is the same definition as the blocker definition but without the min involved. First I will prove that $B(\mathcal{H}^{\min}) = \mathcal{B}$ by showing that $B(\mathcal{H}^{\min}) \subseteq \mathcal{B}$ and that $B(\mathcal{H}^{\min}) \supseteq \mathcal{B}$.

A set of edges $T \subseteq E(G)$ is not in $B(\mathcal{H}^{\min})$ if it does not contain any edges of an odd minimal circuit $c \in \mathcal{G}$ (by definition all circuits are minimal). For T to be in $\mathcal{E}(\mathcal{B})$, one has $E(c) \subseteq \delta(S)$ for an $S \in \mathcal{P}(\mathcal{V})$. That is impossible because S could not have neighboring vertices that are in c . Suppose S contain an arbitrary vertex $v \in E(c)$, then it could not have the neighbors of v but it must have the neighbors of the neighbors of v that are in $E(c)$ because otherwise you would have one missing edge in our desired $\delta(S)$. Pick a direction and start choosing all the necessary vertices using the previous criteria. As c is an odd circuit, at the end you would have to pick a vertex u that is a neighbor of v , which is a contradiction. Given that every non-edge of $B(\mathcal{H}^{\min})$ is a non-edge of $\mathcal{B}$, then $\mathcal{B} \subseteq B(\mathcal{H}^{\min})$.

On the other hand, if $T \subseteq E(G)$ is in $\mathcal{E}(\mathcal{H}^{\min})$, then $G - T$ is a bipartite graph (A, B) because it does not have odd circuits. This is true because for even circuits you can easily have a bipartition (neighboring vertices are in different partitions). If the subgraph is not a circuit you also can have neighboring vertices in different partitions because if you start in the root vertex you will never have an edge that one end is a previous vertex in the path and you will go through all the vertices in the path. Then, if you choose $S = A$, we have $\delta(S) = E(G) \setminus T$. Given that $E(G) \setminus \delta(S) = T$. Thus, $T \in \mathcal{E}(\mathcal{B})$, and therefore $\mathcal{B} \supseteq B(\mathcal{H}^{\min})$.

Given the discussion above $B(\mathcal{H}^{\min}) = \mathcal{B}$. By definition it, is trivial that, if $B(\mathcal{H}^{\min}) = \mathcal{B}$, then $(B(\mathcal{H}^{\min}))^{\min} = \mathcal{B}^{\min}$. Thus, $b(\mathcal{H}^{\min}) = \mathcal{B}^{\min}$.